



EEE283: DIGITAL LOGIC DESIGN
THEORY PROJECT
SECTION: 01
GROUP NO: 12

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1. Project Title:

Safe lock Pro: A Secure room access and automation system

2. Tentative Problem Statement:

In our life security of any object or place is plays a major role. Owing to the improvement and advancement of science and technology over the world there is a subsequent boost in the rate and cleverness of crime and violence by thief and ruffian. As a result, it is necessary to ensure security of oneself and one's valuable belongings. The requirement for a well-equipped home security system is definitely on the rise. Besides to that, home security systems are an important feature of modern residential and office setups. We want to utilize the electronic technology to build a home security system which will be useful in keeping thieves, dacoits and other sort of dangers at bay.

This project has considered about that and created a secure access for a room which needs a password to open the room including password validation, sensors and alarms. The project is anticipated to present a fully functional Secure Room Access System showcasing password validation, access control, sensor integration, and alarm triggers within the Proteus simulation environment. The project seeks to confront real-world security issues, offering valuable insights into embedded systems design,

simulation, and security protocols. The flexibility of Proteus allows for thorough testing, validation, and potential optimizations prior to actual implementation in the real world.

Now a day's most of the systems are automated in order to face new challenges and present day requirements to achieve good results. Automated systems have less manual operations, so that the flexibility, reliabilities are high and accurate. Hence every field prefers automated control systems, especially in the field of electronics.

3. Tentative Objectives:

This project aims to develop an advanced system that integrates secure room access control with automation features for both residential and commercial applications. The system, named Safe lock Pro: A Secure room access and automation system employs cutting-edge technologies to provide a seamless and secure experience for users. Using keypad, it enters a password to the system and if it is entered correctly the user gets access to the room.

The primary challenges of the project involve implementing an effective password validation mechanism, managing unauthorized access attempt counts, seamlessly integrating various components within the Proteus simulation environment, ensuring synchronized responses in diverse

scenarios, and attaining a high degree of realism and reliability in the simulation. The main key tasks to implement here are:

- Develop a password validation system that enables users to enter three digits on a keypad. This password will be compared by the system to a pre-defined password that is kept on file on a server.
- Establish an access control system that only allows entry into the room if the correct password is successfully entered. A green LED will illuminate to clearly convey this, giving the user instant feedback.
- Make the most efficient use of power by turning on the lights and air conditioning only when people are in the room. This way, no electricity is wasted when people are not there.
- Install a security feature that can identify attempts at unauthorized access. As a warning, an alarm will ring if the user enters an incorrect password three times in a row.
- Include a temperature and motion sensor in the system. These sensors will come on after a successful login, allowing the room's surroundings to be monitored.

Here, Password validation, access control, sensor integration, and alarm triggers are things we want to successfully accomplish so the system works as intended in a variety of situations. We want to create a simulation that closely mimics actual circumstances so that the functionality and responses of the system can be tested with accuracy. We want to ensure

security and examine the system's functionality in a variety of situations, such as successful entry following sensor activation, early sensor activation, incorrect password attempts, and successful access.

4. Gantt Chart:

	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15
Research & Brainstorming	28	29	30	1	2	3									
Design and Analysis							4	5	6						
Software simulation										7	8	9			
Completion of report													10	11	12

Group Contribution:

ID	Name	Worked on
21321005	Shahriar Nafi	Tentative Problem Statement, Tentative Objective
21321031	Rifat Bin Reza	Specifications and Requirement, Project Validation
21321035	Mahmood Sayeed Ziyad	Methodology, Results & Discussions
21321041	Khandaker Sazid Ashfaque	Impact, Conclusion and Future work

5. Specifications and Requirements:

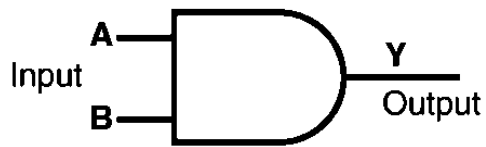
We will design a system that receives a specific input and produces an output signal when the right combination is entered. The underlying specifications and needs for this kind of circuit are listed below:

Specifications:

- Selecting which input-method the combination will be used. It could be a keypad, a set of switches or any other appropriate input device.
- Setting the length of the combination. In our case, a seven-segment binary input display has a two-digit number.
- Specifying what occurs upon entering the correct combination. This could be any desired action. To ascertain whether the identity is correct or incorrect, we have chosen the red and green light.
- Choosing the appropriate response for the circuit when an incorrect combination entry is made. It might include a reset following a predetermined amount of unsuccessful tries

Requirements:

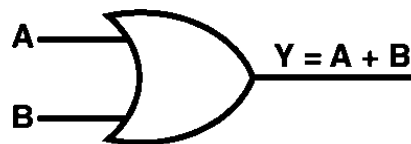
- **AND gate:** In the AND gate, the output of an AND gate attains state 1 if and only if all the inputs are in state 1.



A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Truth Table

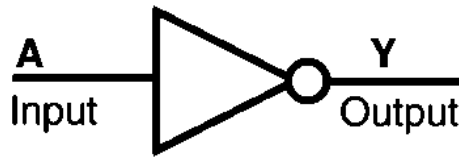
- **OR gate:** In an OR gate, the output of an OR gate attains state 1 if one or more inputs attain state 1.



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Truth Table

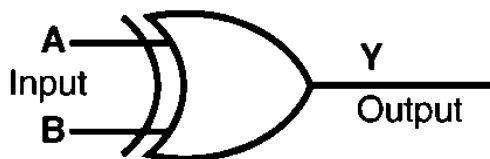
- **NOT gate:** In a NOT gate, the output of a NOT gate attains state 1 if and only if the input does not attain state 1.



A	Y
0	1
1	0

Truth Table

- **XOR gate:** In XOR gate output will be 1 when both inputs are different.

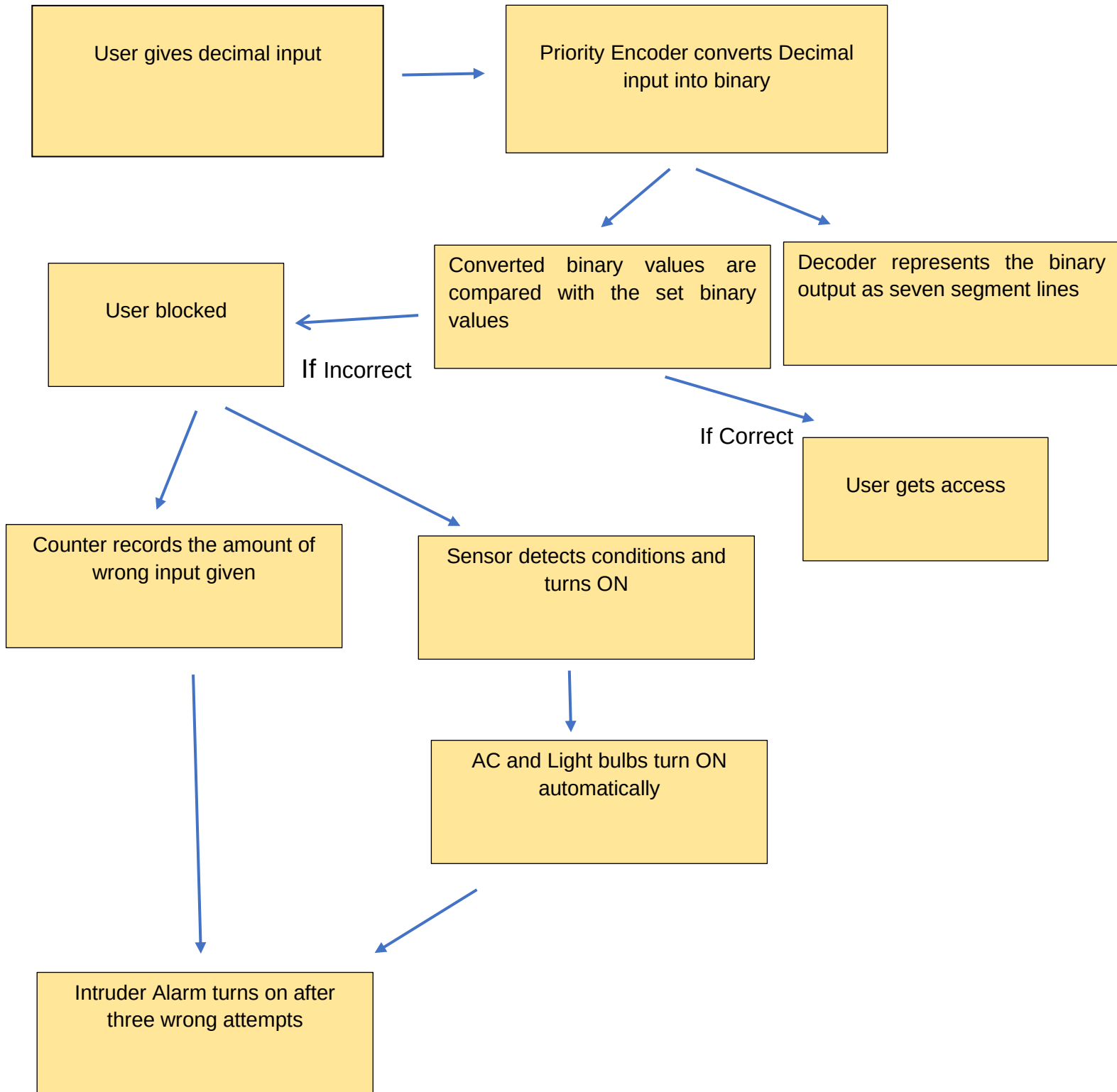


A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Truth Table

- **Flip Flop:** To temporarily store the entered digits and compare them with the correct combination, memory elements are implemented. So, in our case a JK flip flop have been used.
- **Decoder/Encoder:** To convert the decimal input to binary a decoder needs to be used and to convert binary to decimal representation encoder is used.
- **Clock Signal:** Our design involves sequential circuits. So a clock signal is necessary
- **LED:** Light Emitting Diodes are needed so that we can signal whether the password will be correct or not

Flowchart:



Methodology:

For this project, we used a modular design approach. We worked on the various parts separately before integrating them together at the end. This made the normally lengthy and challenging procedure much easier to handle and guaranteed simple troubleshooting at every stage.

Our design steps can be broken down into five segments:

Step 1: Designing a keypad that can print binary data after accepting decimal input

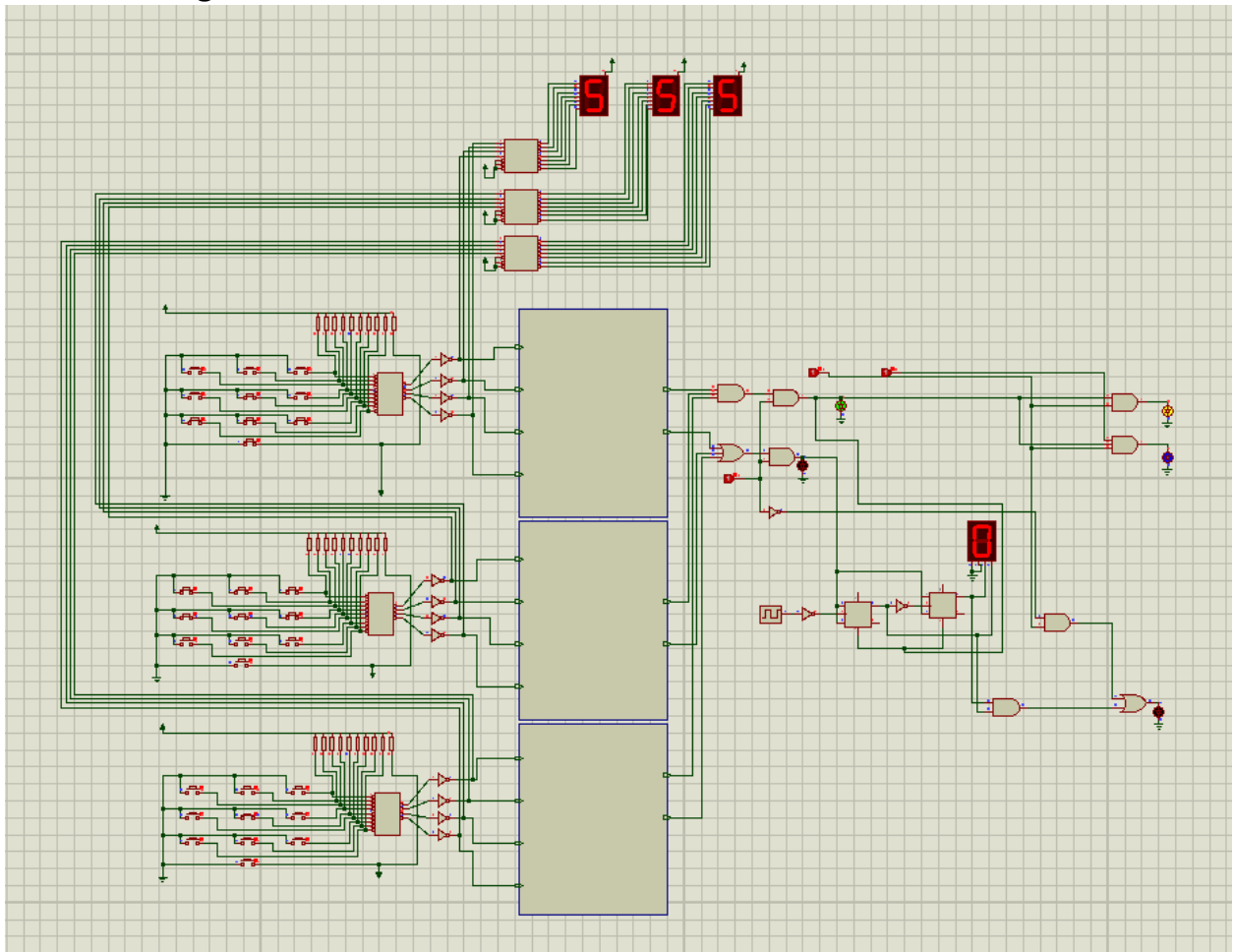
Step 2: A logic circuit that can compare between two BCD inputs

Step 3: A counter to keep track of the number wrong inputs

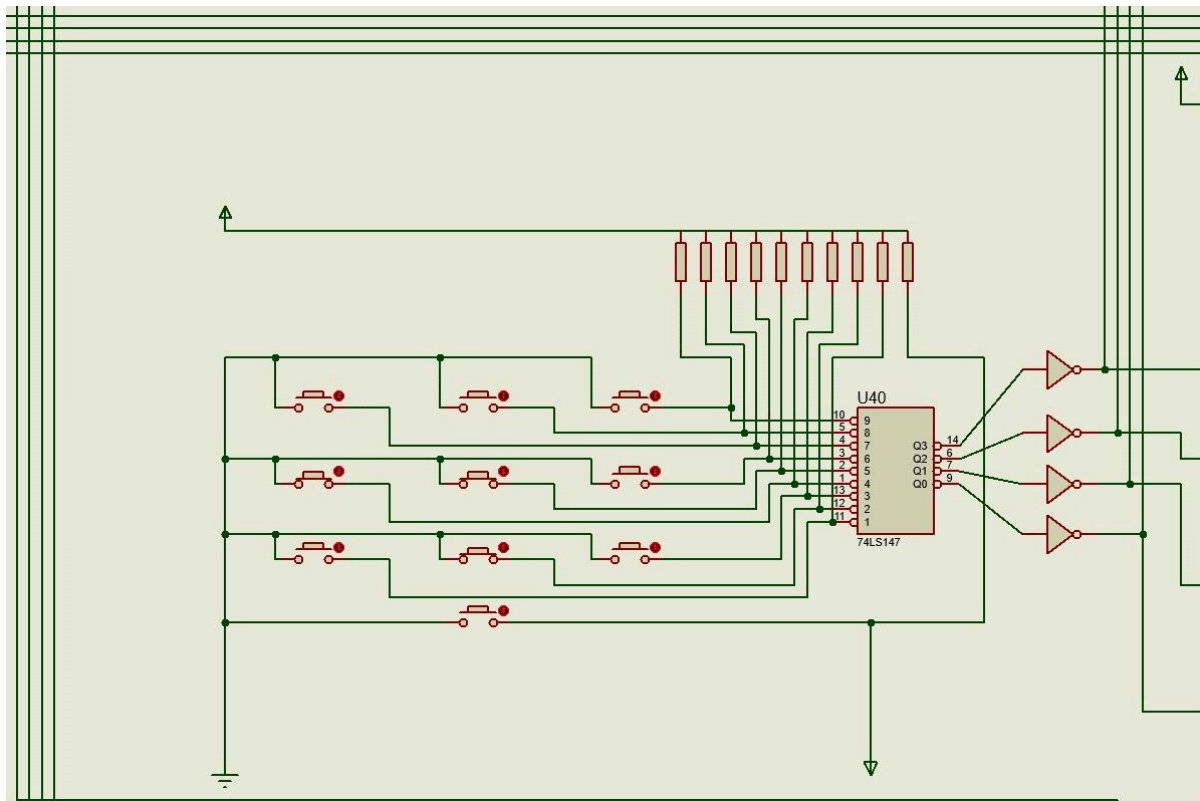
Step 4: Logic circuit for implementing the automated system, using sensors.

Step 5: A feedback mechanism to tackle intruder entries.

Circuit Diagram:



Step 1: Keypad designing which is capable of taking Decimal Input



This consists of 10 push-buttons (0-9). The push-buttons are connected to an encoder which converts decimal input to binary output.

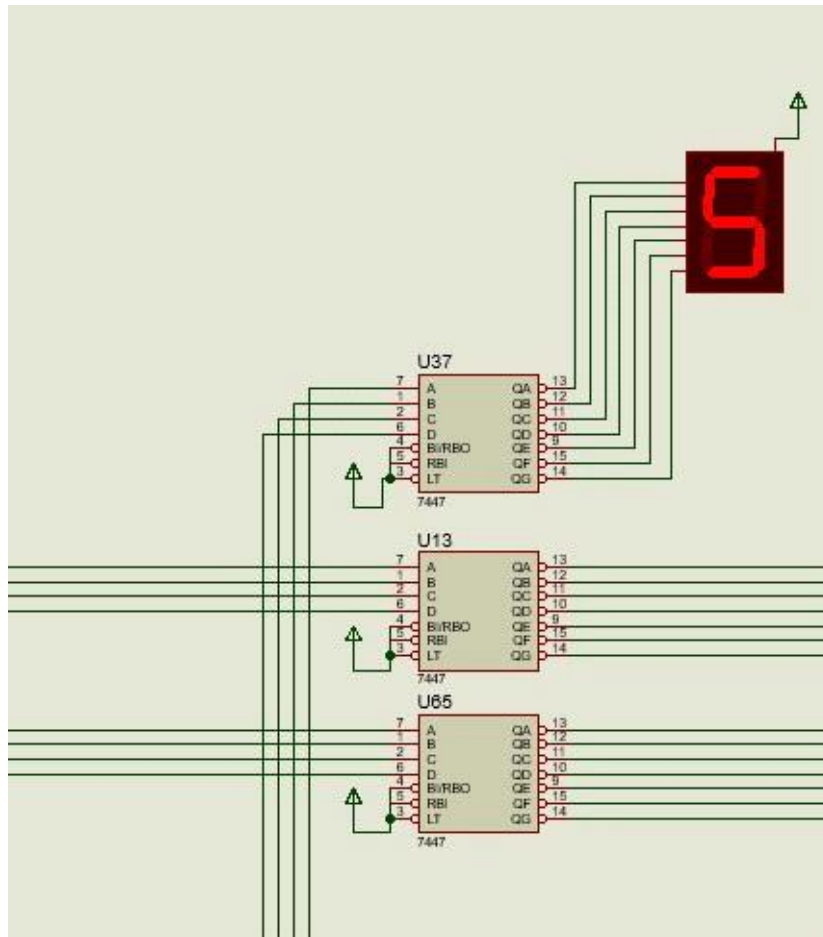
Truth Table:

**SN54/74LS147
FUNCTION TABLE**

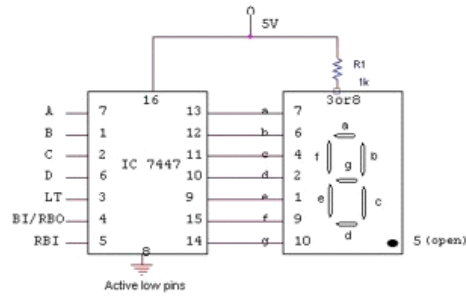
INPUTS									OUTPUTS			
1	2	3	4	5	6	7	8	9	D	C	B	A
H	H	H	H	H	H	H	H	H	H	H	H	H
X	X	X	X	X	X	X	X	L	L	H	H	L
X	X	X	X	X	X	X	L	H	L	H	H	H
X	X	X	X	X	X	L	H	H	H	L	L	L
X	X	X	X	X	L	H	H	H	H	L	L	H
X	X	X	X	L	H	H	H	H	H	L	H	L
X	X	X	L	H	H	H	H	H	H	L	H	H
X	X	L	H	H	H	H	H	H	H	H	L	L
X	L	H	H	H	H	H	H	H	H	H	L	H
L	H	H	H	H	H	H	H	H	H	H	H	L

H = HIGH Logic Level, L = LOW Logic Level, X = Irrelevant

Because the encoder produces low values, each inverter was connected to one binary bit. The purpose of the pull-up resistors was to guarantee that the push buttons were elevated to a voltage sufficient for proper operation. Circuit is thus capable of taking in a single-digit decimal input and gives its equivalent binary output, which is displayed with a connected 7-segment display.



The seven-segment display is connected to a 7447 decoder. The decoder converts the four binary input into segmented lines, which can easily show us the given input.

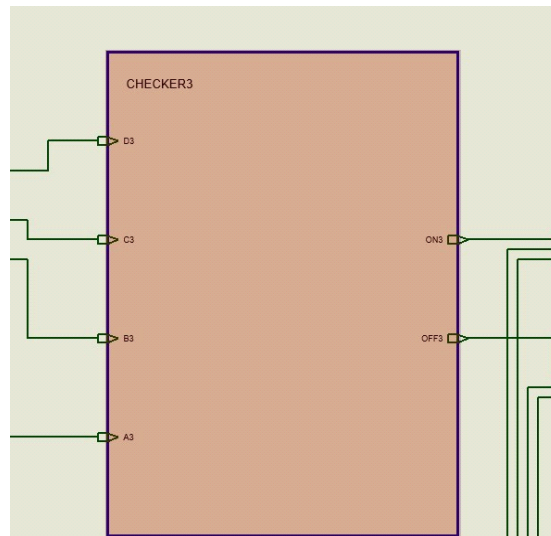


TRUTH TABLE:

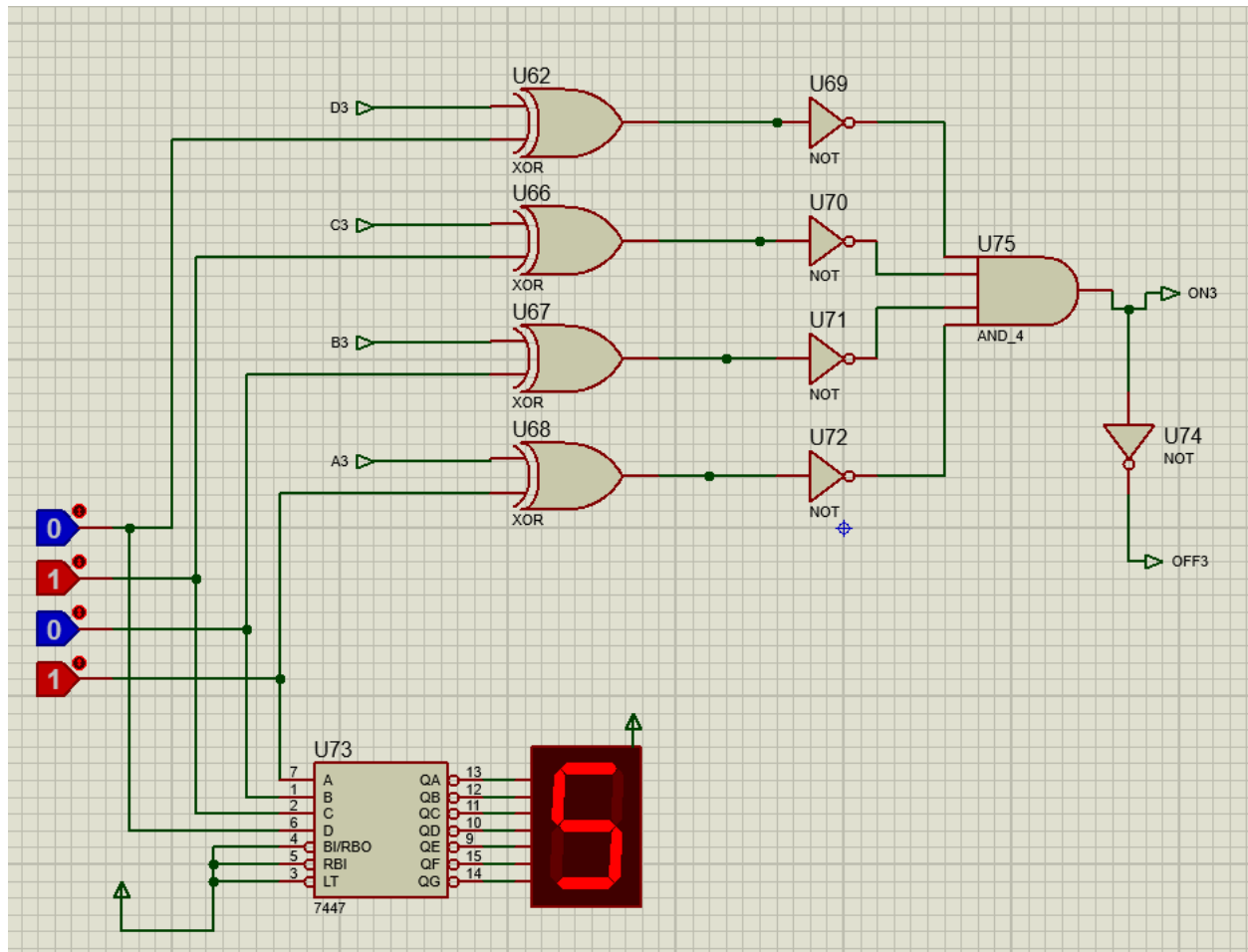
BCD Inputs				Output Logic Levels from IC 7447 to 7-segments							Decimal number display
D	C	B	A	a	b	c	d	e	f	g	
0	0	0	0	0	0	0	0	0	0	1	0
0	0	0	1	1	0	0	1	1	1	1	1
0	0	1	0	0	0	1	0	0	1	0	2
0	0	1	1	0	0	0	0	1	1	0	3
0	1	0	0	1	0	0	1	1	0	0	4
0	1	0	1	0	1	0	0	1	0	0	5
0	1	1	0	1	1	0	0	0	0	0	6
0	1	1	1	0	0	0	1	1	1	1	7
1	0	0	0	0	0	0	0	0	0	0	8
1	0	0	1	0	0	0	1	1	0	0	9

Step 2: Logic Circuit that Can Compare between two BCD

Since the PINCODE is 3-digit long, we have brought the logic circuit capable of comparing between two inputs into three identical sub-circuits:

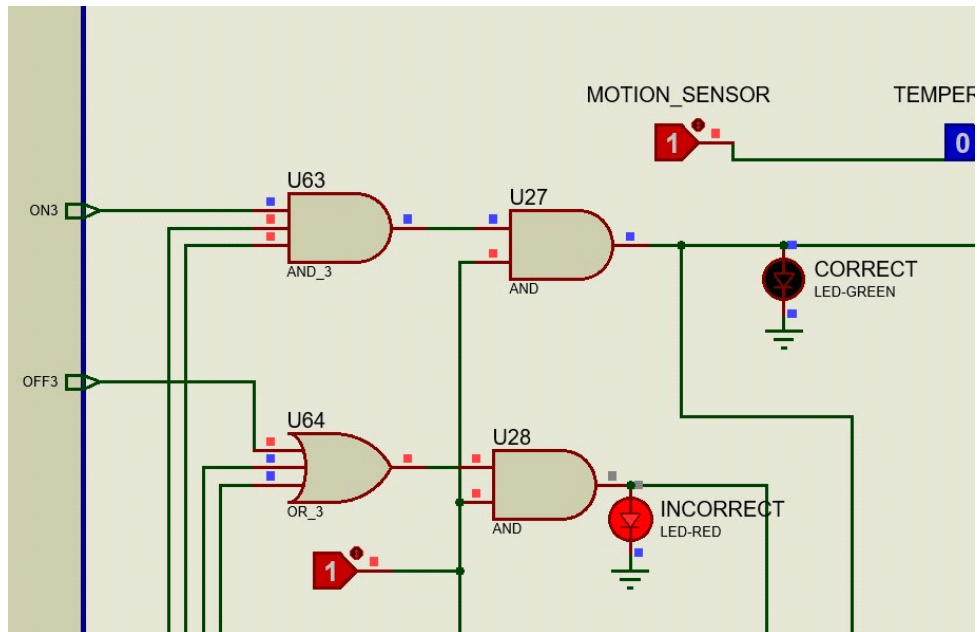


Child sheet of the CHECKER sub-circuit:

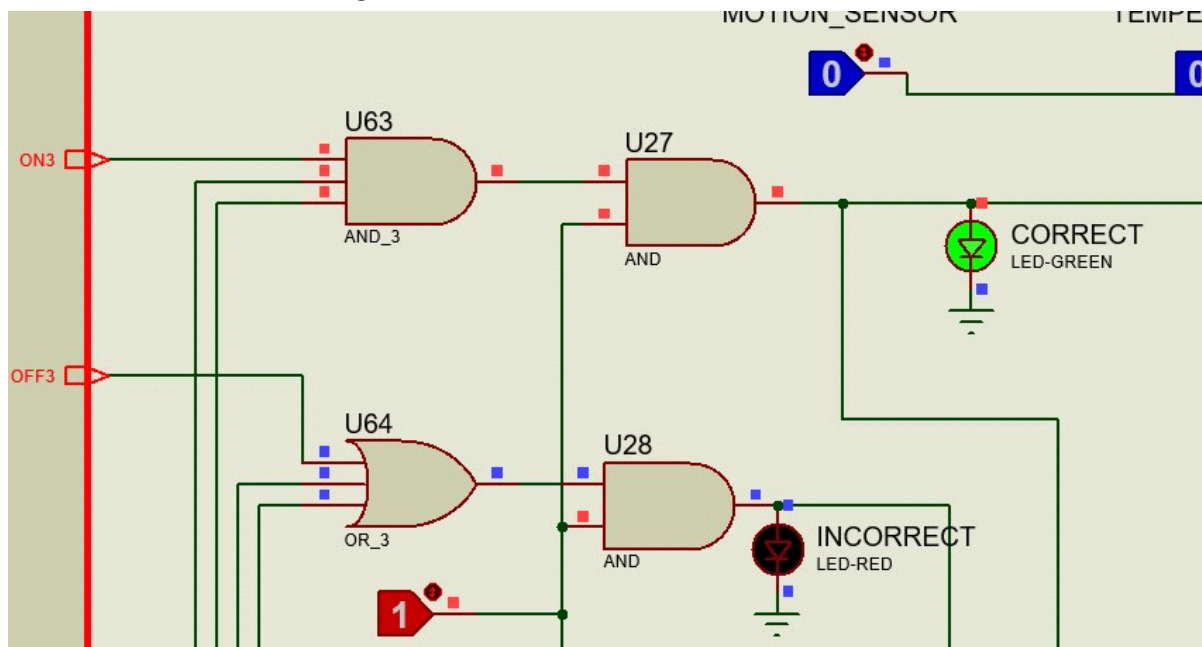


The circuit consists of essentially 4 XOR gates, 4 NOT gate comparing the input bits to stored bits. If all the bits match, 4_input AND turns on, turning on the ON output. If one of them does not match, the OFF output turns on. We have 3 CHECKER sub-circuit for each digit, and all their ON outputs are connected to a 4_input AND gate, hence all the bits of each digit match, and the circuit turns ON. Their OFF outputs are OR_ gated, so if any one of the digits don't match the door won't unlock.

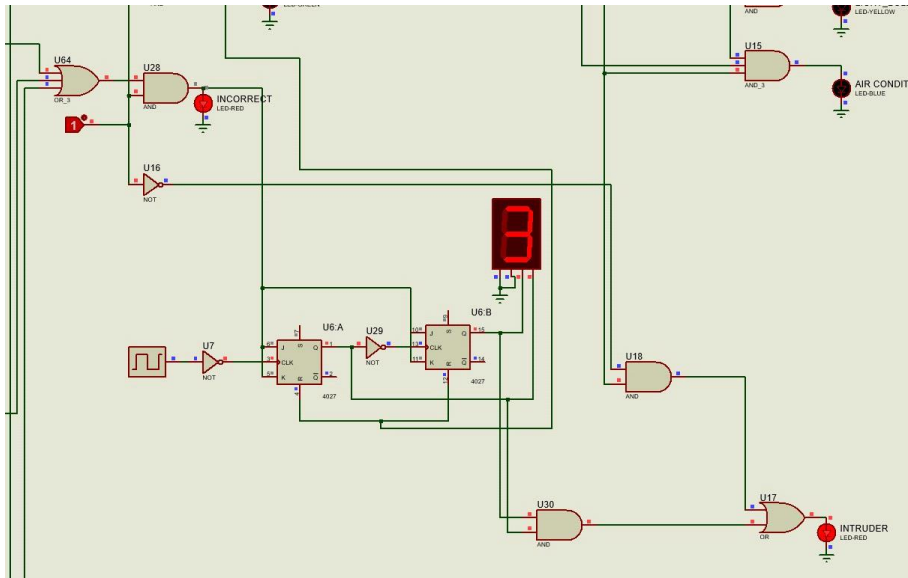
Furthermore, if the authorities so want, they can reset the password bit by bit for each digit by accessing the sub-circuit. To improve system security and deter frequent password resets, it has been purposefully made difficult to retrieve the stored information.



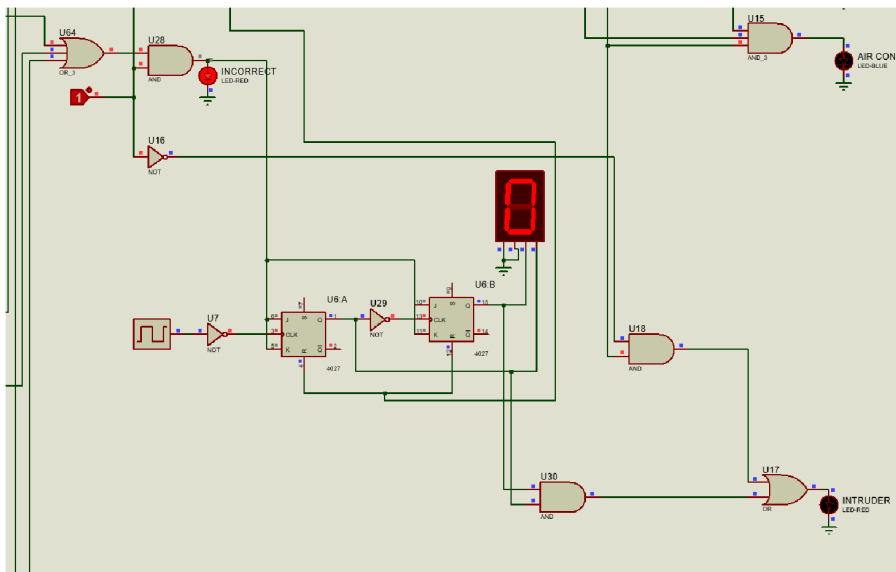
The "ENTER" button functions as a toggle button. The user will touch it to turn on the checker circuit after each PIN entering. To guarantee that the door stays unlocked for the duration of his visit, the user must latch the toggle to "ON" when entering the room.



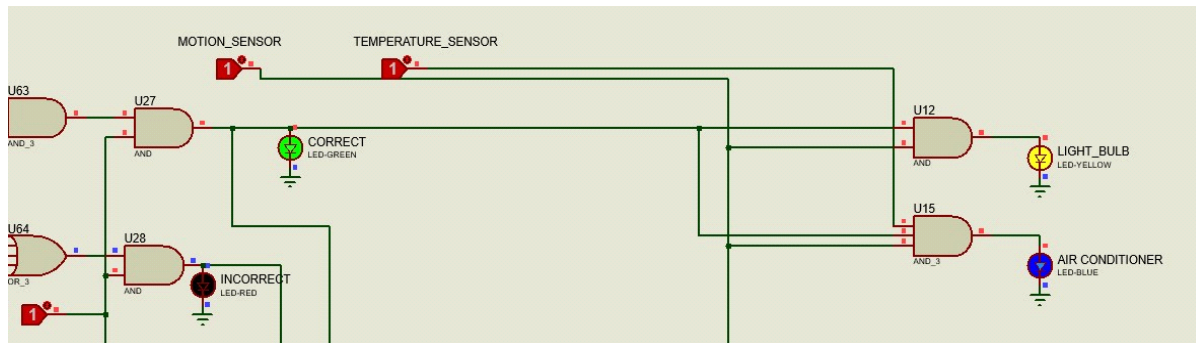
A counter to keep track of the number wrong inputs



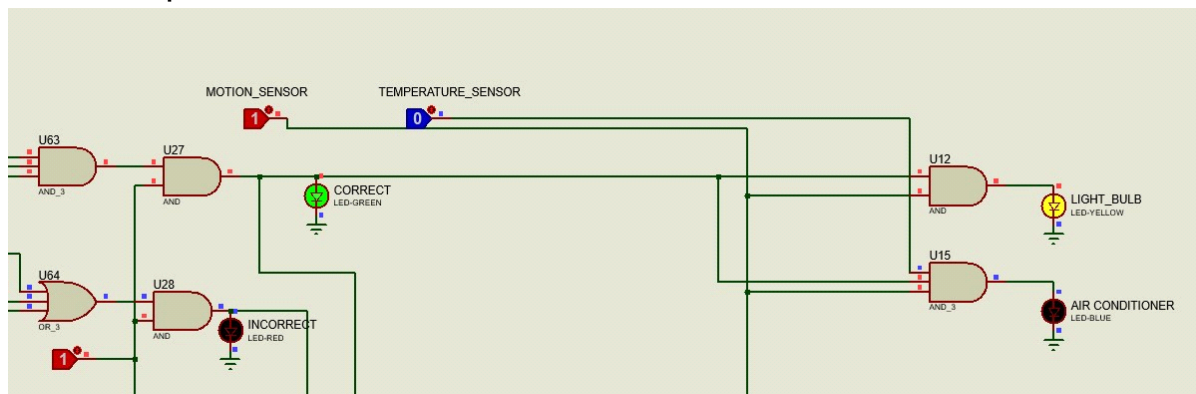
The counter is developed using two J-K flip-flops and it counts up to three before turning on the intruder alarm. A negative edge clock is attached to the flip-flops. (NOT gate makes the clock negative edge). The counter is reset to zero if correct input is given, since the reset is connected to the “CORRECT” pin.



Logic circuit for implementing the automated system, using sensors. The logic circuit for the control of the LIGHT BULB and AIR CONDITIONER has been achieved by implementing the motion sensor, temperature sensor, and AND gates.



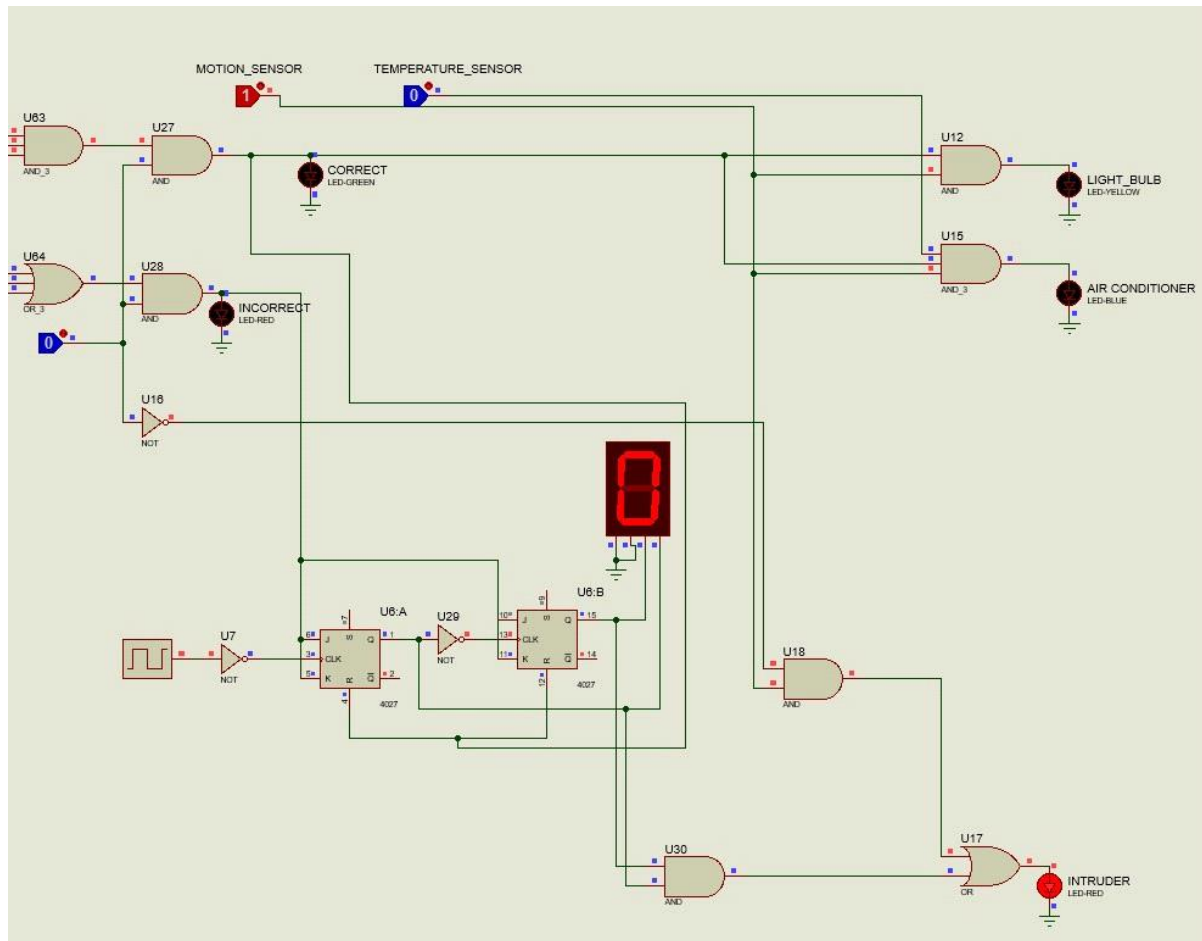
Light bulb turns on if the user gets access to the room, and motion sensor detects a presence:



If the user enters the room, the motion sensor recognises their presence, and the temperature rises over a predetermined point, the air conditioner activates. (set using the temperature sensor):

Feedback mechanism to detect Intruder Entry -

When the motion sensor detects a presence, however no user has been given access to the room, the INTRUDER ALARM turns on. This is achieved by connecting the motion sensor and NOT of the toggle switch to an AND gate .



9. Results & Discussion:

Implementing and simulating the project in the Proteus environment produced encouraging outcomes and perceptive observations. The capabilities and possible uses of the Secure Room Access System, which included password validation, access control, sensor integration, and alarm triggers, were showcased.

Results:

Password Validation: The user's input is successfully compared with the pre-set 3-digit password that is kept on the server via the password validation method. When the password was entered correctly, access was allowed; when it was entered incorrectly, the counter mechanism was triggered.

Access Control The green LED indicator lit up after a successful password enter, clearly verifying that the room had been granted access.

Unauthorized Access Detection: After three consecutive failed password submissions, the alarm replied functionally, successfully discouraging unauthorised attempts to gain access.

Sensor Integration: Following successful access, the temperature and motion sensors successfully activated, demonstrating how the system's environmental monitoring is integrated.

Early Sensor Activation Alarm: The proactive security feature was demonstrated by the system's ability to correctly trigger the alert when the motion or temperature sensors were activated before the right password was entered.

Discussion:

System Robustness: The proactive security feature was demonstrated by the system's ability to correctly trigger the alert when the motion or temperature sensors were activated before the right password was entered.

Realistic Simulation: The Proteus simulation environment provided a useful platform for validating and analysing the reactions of the system, successfully simulating the behaviour of the project components.

Energy Efficiency Potential: Motion sensor integration offers a way to improve energy efficiency by maximising lighting and climate control systems based on occupancy, even though it isn't done so explicitly in this simulation.

Scalability and Practicality: The base of the project in Proteus allows for future development, personalisation, and minimally modified real-world deployment in actual settings.

Interdisciplinary Learning: Through the design and modelling of the project, interdisciplinary knowledge was fostered through the research of embedded

systems, security mechanisms, sensor integration, and energy efficiency principles.

Future Improvements: Future iterations could explore fine-tuning the password validation process, refining sensor activation thresholds, and enhancing user feedback mechanisms.

10. Project Validation: The project aims to design and validate a secure room access system using the Proteus simulation software. The system employs a 3-digit password for room access, activating sensors and lights accordingly. The project demonstrates how to interface a keypad, motion sensor, temperature sensor, LEDs, and an alarm to create a comprehensive security setup.

Functionalities:

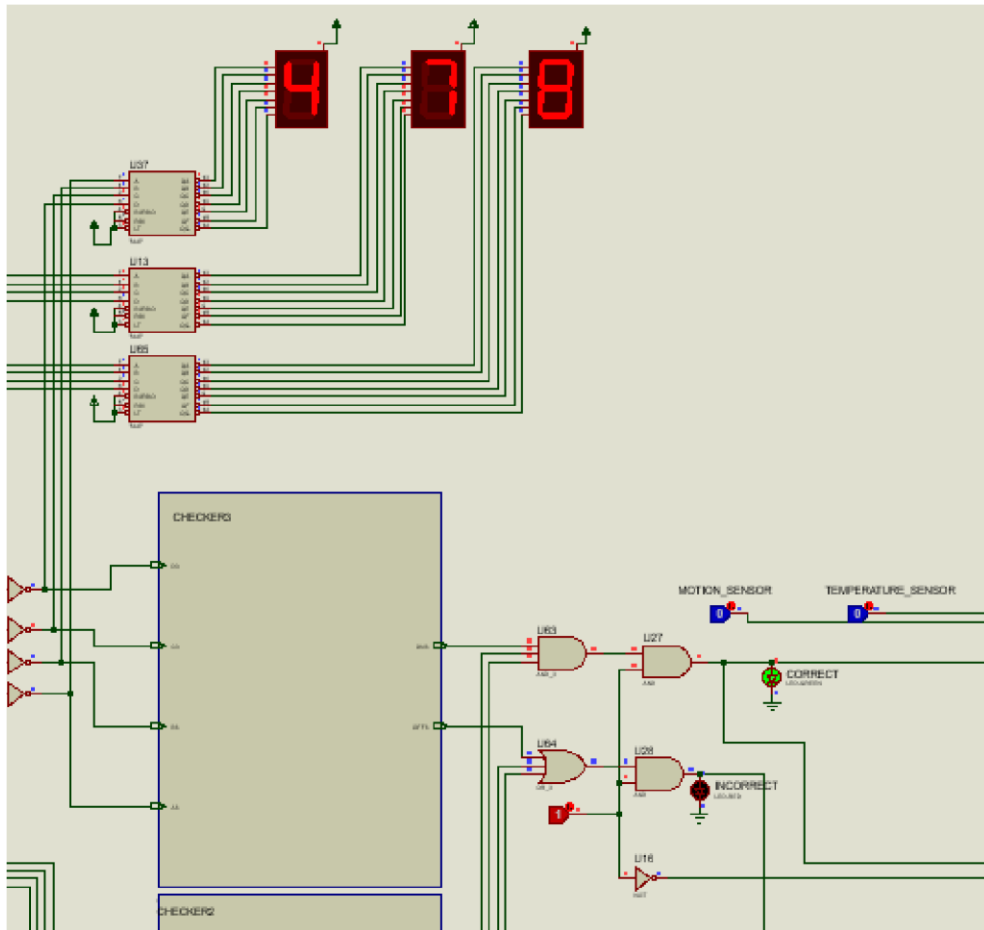
- a) Password Validation:** The system checks if the user-input password matches the predefined password in the server. If the password is correct, access is granted.
- b) Access Control:** After successful password entry, the green LED lights up, indicating access approval.
- c) Sensor Activation:** Upon successful access, the motion sensor and temperature sensor activate, monitoring the room environment.
- d) Unauthorized Attempts:** If the user enters the wrong password three times, the alarm will ring, signaling unauthorized access.
- e) Early Sensor Activation:** If the motion sensor and temperature sensor activate before the correct password is entered, the alarm will sound to indicate a potential breach. The project will demonstrate and vary on different case scenarios. Where the variation of scenarios will result in different output cases.

Case 1: Successful Access

In this scenario, the user correctly enters the 3-digit password, granting access to the room. The validation process compares the user's input with the server-stored password. As the simulation runs:

1. The user interacts with the keypad, entering the correct password.
2. The server validates the password and signals successful access.
3. The green LED lights up, indicating that access has been granted.
4. The motion sensor and temperature sensor activate after the user's successful entry]

Password has been set to 478 -> The user enters correct password -> LED turns green, signaling the user has been given access.

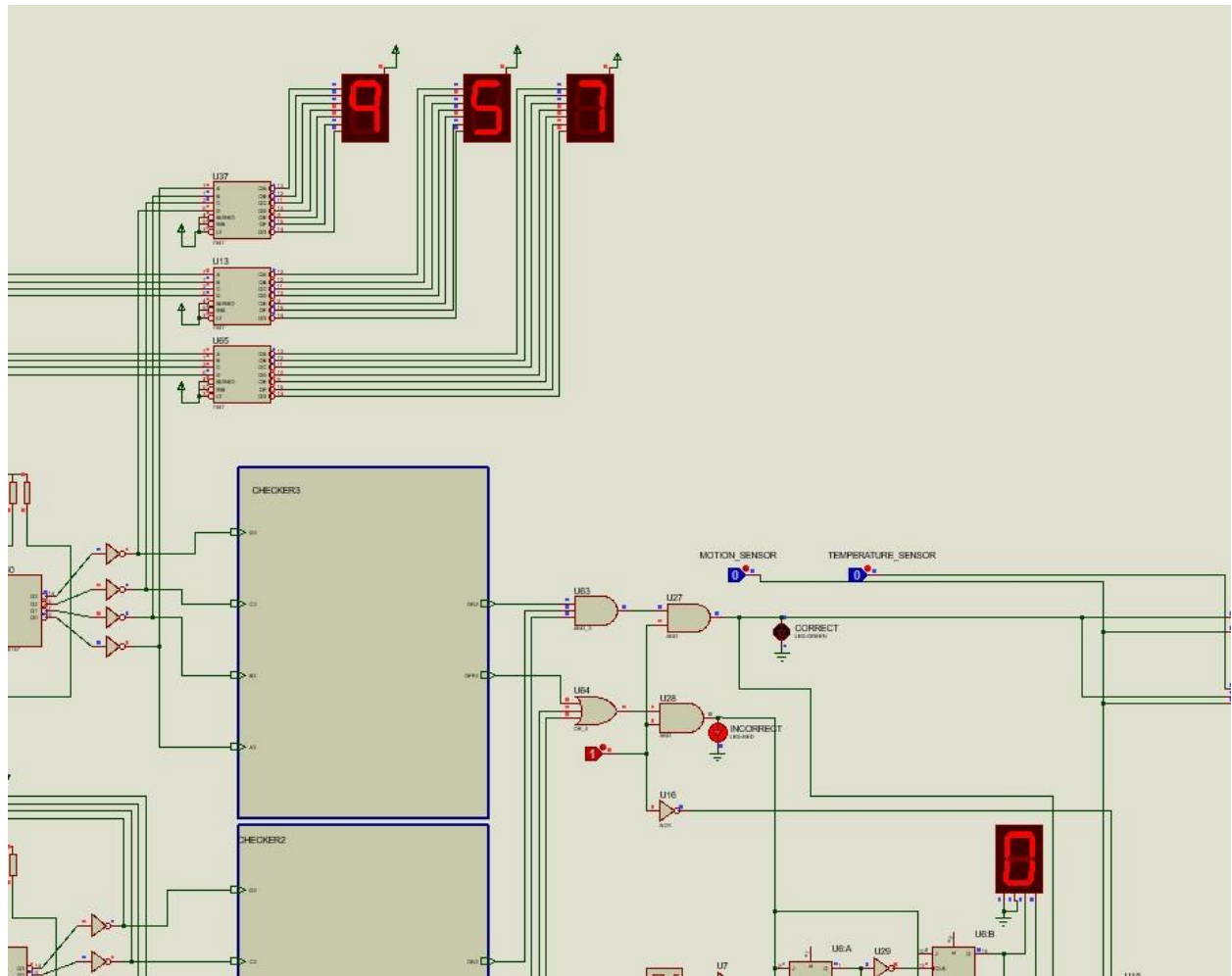


Case 2: Unauthorized Access and Handling Wrong Password Inputs

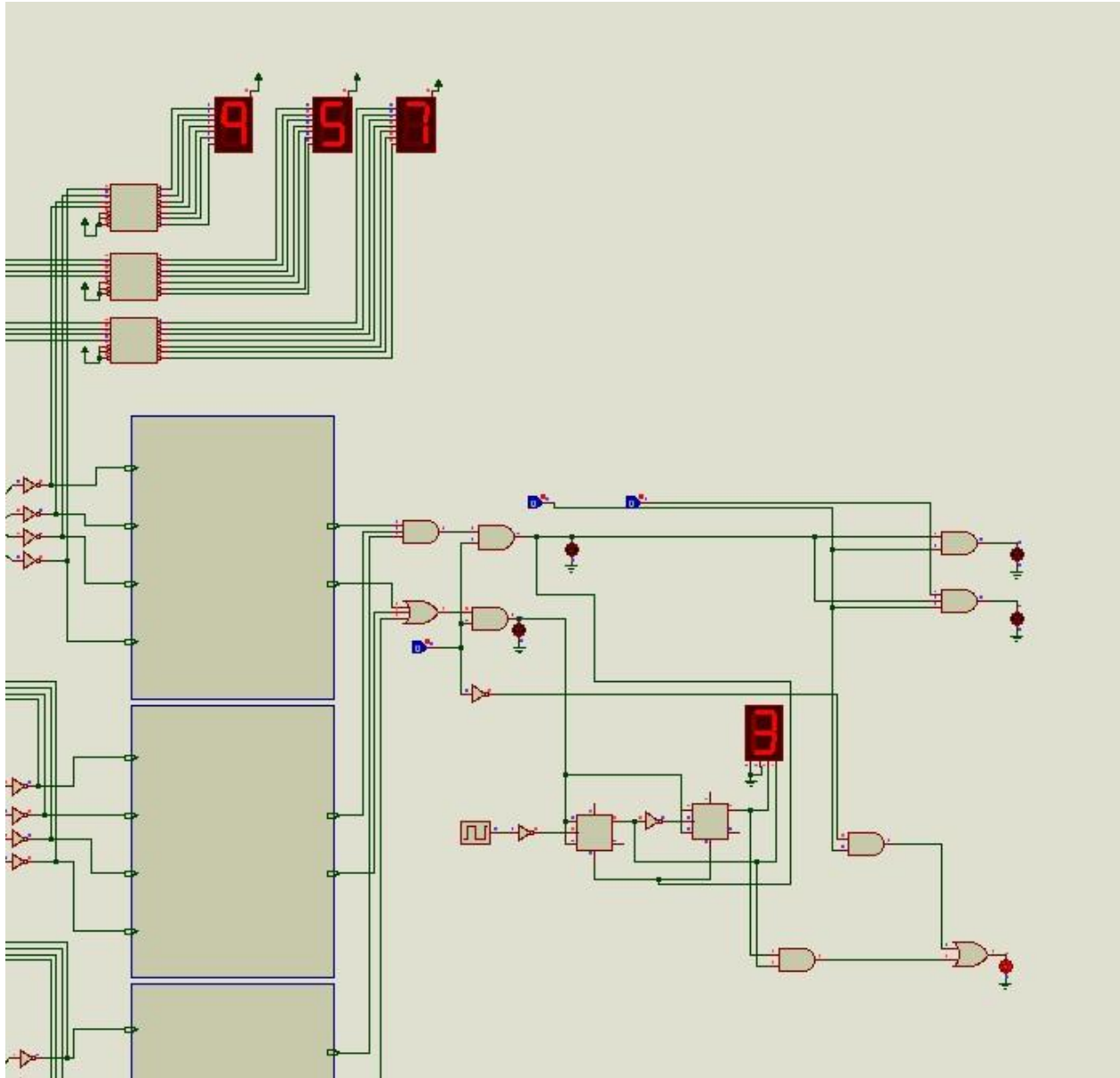
Attempts Here, the user enters the wrong password, access is blocked. If he persists and enters the wrong PIN code three times in a row, an alarm is triggered and security is alerted.

1. The user inputs an incorrect password.
2. The system invalidates the password and blocks access
3. LED turns Red

The password is still 478-> User input the wrong passcode->LED goes red, blocking access.

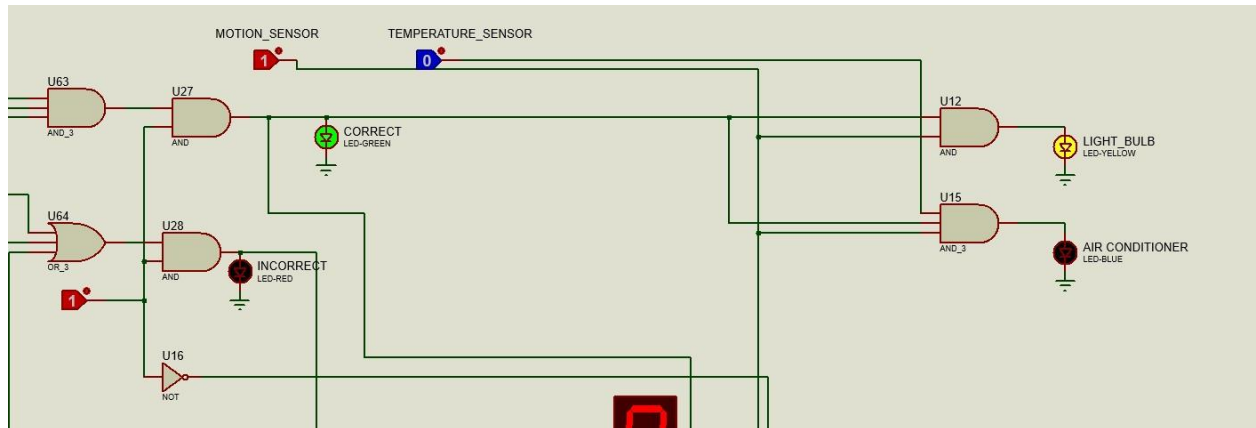


4. The user repeats this process two more times.
5. After the third incorrect attempt, the counter reaches the threshold.
6. The alarm rings, indicating unauthorized access.

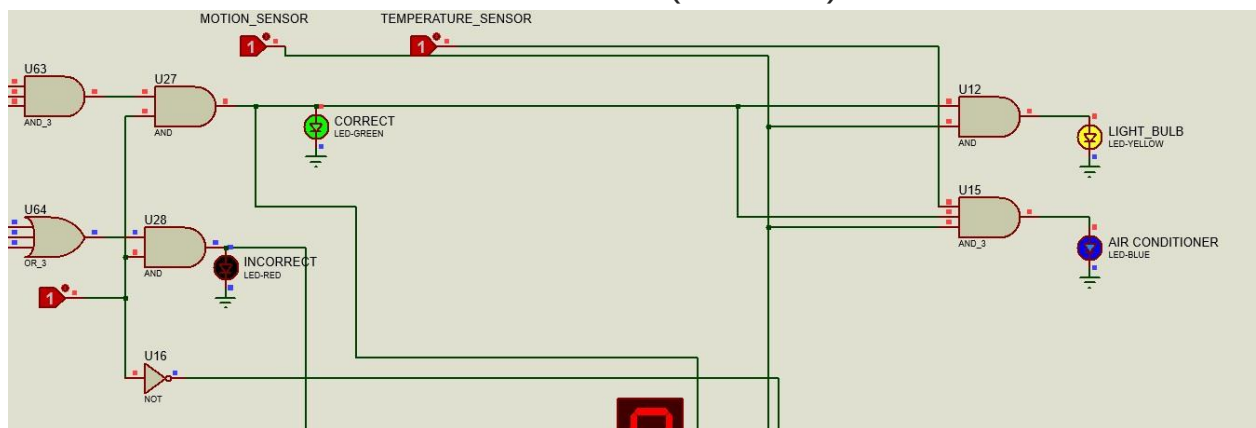


Case 3: Activation of Sensors after successful entry

1. The user enters the room upon successful password entry.
2. The MOTION SENSOR detects his presence and turns ON.
3. LIGHT BULB (Yellow - LED) is turned on.



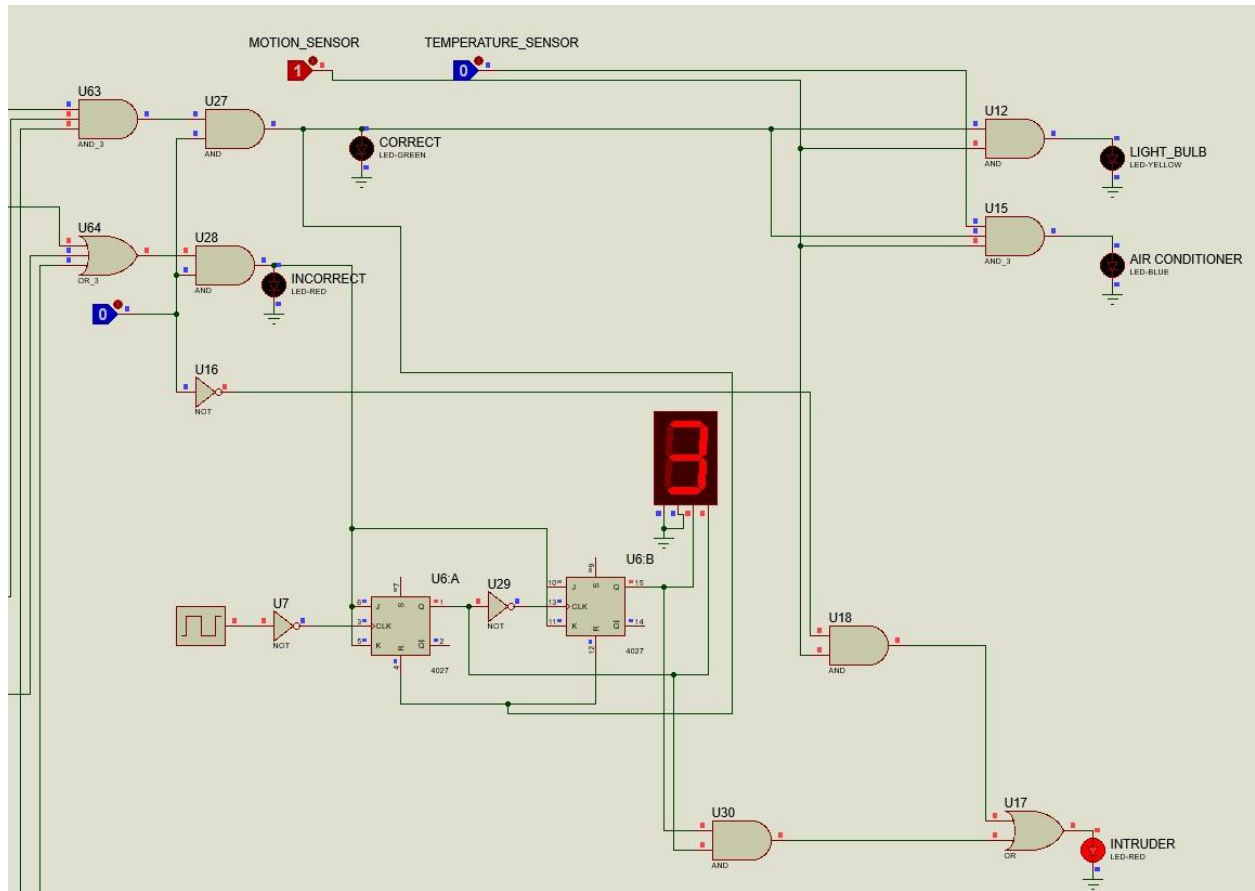
4. If the temperature is above a certain set value, the TEMPERATURE SENSOR turns ON.
5. This causes the AIR CONDITIONER (Led-blue) to turn on as well.



Case 4: Early Sensor Activation with No Password Entry

In this scenario, the motion sensor activates before the user inputs the correct password, causing the alarm to sound due to potential unauthorized access.

1. The motion sensor activates without the server giving access.
2. The system detects the sensor activation before the password is entered.
3. The intruder alarm rings, indicating a potential breach.



Through thorough validation in different scenarios, the secure room access system's functionality has been comprehensively tested within the Proteus simulation environment. The project effectively implements password validation, access control, sensor activation, and alarm triggers. By examining these various scenarios, we have demonstrated the robustness and reliability of the proposed security solution. This project validation underscores the viability of the system's real-world application and provides valuable insights for potential improvements and optimizations.

11. Impact:

Our smart room system aims to revolutionize room access and resource management through advanced technology, offering a multitude of benefits:

- Password-restricted entry: Provides robust access control, ensuring only authorized personnel can enter specific rooms. This significantly reduces the risk of theft and unauthorized use of sensitive equipment or materials.
- Monitoring capabilities: Real-time monitoring helps track room usage and identify any potential security threats, promoting a safe and secure environment.
- Motion detection: Optimizes energy consumption by automatically turning on lights and air conditioning only when valid members are present. This minimizes unnecessary energy usage and promotes a sustainable environment.
- Room-specific access: Eliminates unauthorized entry and ensures resources are used only by designated individuals, further reducing energy waste.
- Room-specific access control: Restricts access to authorized personnel only, preventing unauthorized individuals from entering designated rooms. This safeguards sensitive equipment and materials, promoting a secure and controlled environment.
- Promotes efficient resource utilization: Enhances resource allocation by ensuring only authorized personnel have access to specific rooms and equipment, preventing misuse and unnecessary waste.
- Controlled access: Prevents overcrowding in rooms, ensuring a comfortable and productive environment for authorized users.
- Increased transparency: Allows users to view room availability in real-time, enabling them to plan their time effectively and avoid unnecessary waiting times.

- **Guaranteed access:** Provides equal access to lab resources for all authorized students, fostering fairness and eliminating access-related grievances.
- **Promotes efficient resource utilization:** Ensures all students have the opportunity to utilize available resources, maximizing their learning and research potential.
- **Real-time monitoring:** Provides valuable insights for personnel responsible for room supervision, allowing them to effectively track room usage and identify any potential issues.
- **Improved accountability:** Supports personnel in managing large groups of students and facilitates efficient resource allocation.
- **Minimal maintenance:** Designed for long-term operation with minimal maintenance requirements, reducing operational costs and staff workload.
- **Increased efficiency:** Automates room access and resource management, freeing up valuable time and resources for other tasks.

By implementing this smart room system, we can achieve enhanced security, optimize resource utilization, and create a more efficient and sustainable environment for all users.

12. Conclusion

The widespread use of open access rooms in educational institutions, particularly labs and classrooms, has led to significant challenges in managing security, privacy, energy consumption, and access control. Overcrowding, unequal access, and power wastage are just a few of the pressing concerns associated with these spaces.

Our innovative solution, the Secure Room Access System, addresses these issues by providing a comprehensive and automated access control system for educational institutions. This system offers a multifaceted solution that:

- Enhances security: By implementing password-restricted entry and real-time monitoring, the system significantly reduces the risk of unauthorized access, theft, and misuse of equipment and information.
- Optimizes resource utilization: Motion detection technology automatically triggers lights and air conditioning only when authorized personnel are present, minimizing energy waste and promoting sustainability.
- Guarantees proper access: Room-specific access control ensures that designated resources are utilized only by authorized students, fostering fairness and eliminating issues related to unequal access.

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This comprehensive solution not only enhances security and optimizes resource utilization but also fosters a more productive and equitable learning environment for students. By ensuring proper access and dedicated room time, the system ultimately contributes to higher educational quality and improved student outcomes.

The successful simulation of the Secure Room Access System within the Proteus environment validates its viability as a comprehensive solution for educational institutions. This project exemplifies the convergence of security and technology, demonstrating the potential for efficient access control and energy-saving measures. The results and insights obtained from this simulation provide a strong foundation for potential real-world implementation and offer valuable learnings for the realms of security, energy efficiency, and embedded systems design.

13. Future Work:

While the immediate benefits of the smart room system are evident in educational settings, its potential extends far beyond university walls. The system's modular design and adaptability allow for its seamless integration into various open-room environments, such as corporate offices and even residential households. This universal applicability opens doors to broader adoption and impact.

Furthermore, the environmentally conscious design, featuring energy-saving motion sensors, can be expanded to other infrastructure, including staircases, street lamps, and bathrooms. This green initiative can significantly contribute to energy conservation, especially in areas facing energy challenges.

Finally, incorporating data analytics into the system opens up exciting possibilities. By analyzing power usage patterns and identifying peak usage periods, the system can further optimize resource allocation and maximize efficiency. This data-driven approach can further enhance the system's value proposition and broaden its reach.